INCH-POUND

MIL-R-52968D(ME)
22 November 1988
SUPERSEDING
MIL-R-52968C(ME)
20 August 1982

MILITARY SPECIFICATION

RAILWAY CAR, FLAT, 56-1/2-INCH GAGE, HEAVY DUTY, 140-TON CAPACITY, DOMESTIC SERVICE

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers heavy duty flat cars for transporting tracked vehicles, and American National Standards Institute/International Organization for Standardization (ANSI/ISO) containers.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards, form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-E-489 - Enamel, Alkyd, Gloss, (For Exterior and Interior Surfaces).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-3558	- Plate, Identification, Locomotive, Railway Cars, and Work Equipment.
DOD-C-24667	- Coating System, Nonskid, For Roll Or Spray Application.
MIL-P-52192	- Primer Coating, Epoxy.
MIL-P-53044	- Painting and Marking: Freight and Maintenance Cars Railway Motive Power and Work Equipment.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

ME

13226E7053	- Coupling Link.
13226E7054	- Claw Hook.
13226E7055	- Sling Hook.
13226E7056	- Shackle
13226E7057	- Master Ring.

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-VO, Fort Belvoir, VA 22060-5606.)

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Railroad Administration, Office of Safety.

Railroad Safety Appliance Standards, Part 231 (Title 49 of Code of Federal Regulations).

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(Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (sss 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY OF TESTING AND MATERIALS (ASTM)

A53 - Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

D15.1 Railroad Welding Specification

(Application for copies should be addressed to the American Welding Society, 2501, NW 7th Street, Miami, FL 33125.)

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Manual of Standards and Recommended Practices, Mechanical Division

Section B - Couplers and Freight Car Draft Components.
M-921 Special Cushioning Devices for Freight Cars.

Section C, Part II, Volume I - Specifications for Design, Fabrication and Construction of Freight Cars, M-1001.

Section D - Trucks and Truck Details.
M-114 Helical Springs, Heat Treated Steel.

Section E - Brakes and Brake Equipment.
S-400 Specification No. 2518, Freight Car Brake Equipment, Installation Specification.
S-401 Basic Freight Car Design Data.
S-473 Application Form for AAR Approval of New and Untried Freight Cars.
M-601 Hose, Air-Brake and Train Air-Signal.

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Section E, Part II - Maintenance Requirements for Freight Car Air Brake Control.

Valves and Equipment - Shop Certification.
S-486 Single Car Testing Device - Code of Test for Freight Car Equipment. Instruction Pamphlet No. 5039-4, Sup. 1.

Section G - Wheels and Axles.
M-101 Axles, Carbon Steel, Non-Heat-Treated and Heat-Treated.

Section G, Part II - Wheel and Axle (Shop) Manual.

Section H - Journal Bearings and Lubrication.
M-934 Freight Car Journal Roller Bearings.
S-701 Locking Plate, Journal Roller Bearings.

Section I - Specially Equipped Freight Car and Intermodal Equipment.
M-952 Intermodal Support And Securement System for Freight Cars.

Section L - Lettering and Marking of Cars.

Supplement to Manual of Standards and Recommended Practices.
Equipment Diagram, AAR Plates B and B-1 for Unrestricted Interchange Service.

Interchange Rules
Loading of Commodities on Open Top Cars, Section No. 1 - General Rules.

(Application for copies should be addressed to the Association of American Railroads, 50 F Street, N.W., Washington, D.C. 20001.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Description. The flat car shall be 140 ton nominal capacity, 12 wheels, 56-1/2-inch gage in accordance with the applicable Association of American Railroads (AAR) requirements and as specified herein. The flat car shall be designed to carry two M1 tanks, multiple ANSI/ISO Standard containers to be loaded with class A explosives and other commodities. It shall be equipped with

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integral securement systems to restrain each of these items. The car shall be designed to permit movement of tracked vehicles from car to car without the use of bridge plates. The underframe shall be welded construction. Cars shall be designed and constructed in accordance with AAR specifications.

3.1.1 Overall dimensions. The flat car shall be designed to meet these overall dimensions:

Minimum width at bolsters (see 3.10)	10'-5"
Minimum width at endsills	Max. allow. to meet plates B & B-1 requirements
Light car deck height (±1")	49"
Nominal distance between truck centers	46'-3"
Nominal wheel base between axles	5'-6"
Nominal length of car over striker plates	68'-2"
Maximum span between like cars (in draft position)	60"

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall conform to applicable AAR and commercial specifications currently used by the railroad industry for this type of equipment unless otherwise specified herein.

3.3.1 Material deterioration prevention and control. The car shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the car may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Threaded parts. Threaded parts shall conform to FED-STD-H28. The American National fine thread series shall be used for threaded parts less than 1/4-inch diameter. The Unified or American National coarse thread series shall be used for threaded parts 1/4-inch diameter and larger; however, the American National fine thread series may be employed for these sizes when applicable. Airbrake pipe threads shall conform to the American National pipe thread series.

3.3.5 Substitution of materials. Materials not conforming to the specifications referenced herein shall not be substituted, unless such substitutions are specifically approved by the contracting officer and; (1) it is clearly demonstrated that an improvement in operating characteristics, a saving in weight, conservation of critical or strategic materials, or a reduction in

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cost without sacrifice of reliability can be accomplished thereby; (2) that such substitutions will not preclude the subsequent use of specified materials in effecting repairs or replacements. Requests for approval of such substitutions shall be accompanied by completely detailed supporting data which clearly illustrate the use of the proposed substitute material.

* 3.3.6 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the railway car may be newly fabricated from recovered materials to the maximum extent practicable, provided the railway car produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the railway car.

3.4 Interchangeability. Jigs, templates, gauges, and fixtures shall be used to insure interchangeability of like components and parts with respect to assembly, performance and quality. All parts having the same part number shall be functionally and dimensionally interchangeable.

3.5 Tolerances and fits. Tolerances and fits shall conform to appropriate AAR Manuals applicable to railway cars and as specified herein.

3.5.1 Adjustments. All parts and components subject to adjustment or replacement by operating or maintenance personnel shall be accessible. Adjusting facilities where provided shall operate without interference and shall maintain such adjustments under all conditions of service.

3.6 Clearance. Clearance dimensions shall not exceed those shown on AAR Plates B and B-1 for unrestricted interchange service. Maximum allowable width is required for supporting tracks of the M1 tank.

3.7. Curvature. The car shall be capable of negotiating a 38-degree horizontal curve (150-foot radius) when uncoupled. The car shall be capable of negotiating a 150-foot radius horizontal curve when coupled by an auxiliary type coupler similar to National 31629 (see 4.5.2.6). Car shall be designed to negotiate specified horizontal and AAR required vertical curves without interference between trucks and carbody, attached parts or brake riggings. The design shall include allowance for truck springs deflected to 75 percent of their total travel plus body deflection.

3.8 Design and construction. Design and construction shall be in accordance with the AAR Manual of Standards and Recommended Practices, Section C, part II Volume I - Specifications for Design, Fabrication and Construction of Freight Cars, M-1001, and AAR supplement to the Manual of Standards and Recommended Practices, except as specified herein. The car shall be designed to carry the maximum allowable payload consistent with AAR Interchange Rule 89. Structural members shall be free of notch sensitive areas and discontinuities that could result in highly localized stress concentrations. If the design is determined by

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the AAR to be of the "new and untried" type, all requirements to gain AAR interchange approval must be met. A duplicate set of all correspondence, drawings, and design calculations furnished to the AAR shall be submitted to the contracting officer.

3.8.1 Jacking pads and lifting lugs. Car shall be equipped with jacking pads and lifting lugs in accordance with AAR specifications.

3.9 Underframe. The underframe shall be welded construction in accordance with AAR Manual of Standards and Recommended Practices and Supplement thereto. The underframe shall be designed to sustain 600,000 pounds of coupler force applied at the end of the car. The center sill, rear draft stops and their connections shall be designed to withstand a static compressive end load of 1,000,000 pounds.

3.10 Decking. Deck floor plates shall be welded to the underframe and shall extend the full width of the car (3.1.1). For anti-skid protection, a special compound shall be properly applied (see 3.17.1).

* 3.11 Cargo securement system. The securement system shall be capable of containing the maximum allowable payload while undergoing AAR rail impact tests in accordance with part 3, section no. 1 of the AAR loading rules without failure, damage or permanent deformation. The securement system shall also be capable of withstanding without failure, transportation vibration equivalent to that produced by over-the-road movement in a 150-car train. The securement system shall be self-contained on the car deck, shall provide sufficient space for storage below deck level when not in use, and shall not require periodic maintenance, lubrication or adjustment to maintain its serviceability.

3.11.1 Tank securement system. The securement system shall be of the latest AAR approved 48 chain tiedown system with end hook configuration. The system shall consist of these components: Steel anchoring channels, coupling link conforming to 13226E7053, load binder, compression unit, coupling link (same as above), claw hook conforming to 13226E7054, 1/2-inch alloy steel chain, and a sling hook conforming to 13226E7055. Four extra complete tiedown assemblies to be used as spares shall be provided with each car. Four channels shall be flush-mounted parallel along the entire length of the car. The bolster area may be excluded. The heavy-duty anchoring devices shall move freely in the channels, with the ability to be locked at positions in maximum of three-inch increments along the full length of the channels. The anchoring device shall remain in the channels when not in use. The securement devices shall be designed to have a minimum breaking strength of not less than 50,000 pounds. The construction shall be such that one man can position anchors and chain assembly without special tools. Each chain tie-down and anchoring device shall be permanently fixed to the flatcar. A complete set of shackles and masterlinks conforming to 13226E7056 and 13226E7057 and sufficient to secure two M1 tanks to the flatcar shall also be provided with the car.

3.11.2 Intermodal containers. The securement shall be either of the built-in twist lock type or pedestal type for securing a single ANSI/ISO standard size 40-foot-long container; three 20-foot containers or combinations of both. The twist

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lock type shall adjust vertically for different container locking heights, and shall retract flush with the floor level by use of single operating handles when not in use. The built-in pedestal type shall automatically lock the container into position when the container is lowered onto the pedestals. Upward lift of the container shall automatically release the lock for unloading. Construction shall be such that the operator can position the pedestal without special tools. The pedestals shall be permanently secured to the flat car by means of an anti-theft device. The securement system shall meet requirements of the AAR Manual of Standards and Recommended Practices, section I, Specially Equipped Freight Cars and Intermodal Equipment, M-952, Intermodal Support and Securement System for Freight Cars.

3.12 Couplers. Couplers shall be AAR approved bottom shelf E type, rotary bottom operated with coupler operating levers at diagonal corners. Design and installation shall be in conformance with the AAR Manual of Standards and Recommended Practices, Section B - Couplers and Freight Car Draft Components, as applicable.

3.12.1 Cushioning device. The car shall be equipped with end-of-car hydraulic cushioning devices as approved in AAR Specification M-921, Special Cushioning Devices for Freight Cars. The cushioning device shall have a 15-inch travel. Design and installation shall be in accordance with AAR Manual of Standards and Recommended Practices, Section E - Brakes and Brake Equipment; S-400 Specification No. 2518, Freight Car Brake Equipment, Installation Specifications; S-401, Basic Freight Car Design Data; and other AAR requirements as applicable.

3.13.1 Airbrakes. Airbrakes, single capacity, providing constant braking force, shall be furnished and shall consist of an AB-10-12 brake system with ABDW control valve and approved vent valve or an appropriately-sized truck mounted brake system. High friction composition brake shoes shall be in accordance with latest AAR requirements at time of solicitation. The car brake piping system shall consist of 1-1/4 inch extra heavy pipe conforming to ASTM A-53 or latest revision and shall meet the requirements of AAR. Brake valves and breathers shall be protected from damaging effect of road splash and passing couplers. Cars shall pass testing as specified in AAR Standard S-486 Single Car Testing Device - Code of Tests for Freight Car Equipment, Instruction Pamphlet No. 5039-4, Sup. 1.

3.13.2 Braking force. Net airbrake and handbrake force, as determined by a static brake test shall conform to AAR Standards S-401 as above, and S-473 Application Form for AAR approval of new and untried freight cars.

3.13.3 Airbrake hose. Airbrake hose shall conform to AAR Specification M-601 Hose, Air-Brake and Train Air-Signal.

3.13.4 Handbrakes. The handbrake shall be side-mounted below deck level. It shall be operable from deck and ground level.

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3.13.5 Badge plate. A metal badge plate, cast, etched, or stamped, showing brake rigging and dimensions as specified in AAR Manual of Standards and Recommended Practices, section D - Trucks and Truck Details, shall be attached in an accessible location near each applicable system.

3.14 Trucks. Trucks shall be six-wheel, 56-1/2 inch gage designed for a car capacity as required by 3.8 and 3.14.3, and shall be an AAR approved design in accordance with Section D - Trucks and Truck Details. Snubbing device shall be an AAR approved design.

3.14.1 Wheels. Wheels shall be 36-inch one-wear type cast or wrought steel, as specified in the AAR's section G - Wheels and Axles and shall be in accordance with the aforementioned section and other applicable AAR requirements. Wheels shall be inspected and mounted in accordance with AAR's section G, part II - Wheel and Axle (Shop) Manual.

3.14.2 Axles. Axles shall be forged carbon steel in accordance with AAR requirements. Ends of axles shall be drilled and tapped for roller bearing retaining caps.

3.14.3 Journal roller bearings. Journal roller bearings shall be no field lubrication (NFL) type with 6-1/2-inch x 12-inch journal conforming to AAR Specification M-934 Freight Car Journal Roller Bearings in section H - Journal Bearings and Lubrication, and other AAR requirements as applicable.

3.14.3.1 Locking plate. Each roller bearing assembly shall include a locking plate in accordance with AAR Standard S-701 Locking Plate, Journal Roller Bearings.

3.14.4 Springs. Springs shall conform to AAR Specification M-114 Helical Springs, Heat Treated Steel. The spring travel shall be compatible with snubbing device used. The solid spring group capacity shall conform to AAR requirements.

3.15 Side bearings. Each car shall be equipped with AAR approved friction or double roller-type side bearings adjusted to AAR specifications.

3.16 Identification plates. An identification plate conforming to MIL-P-3558, type II, shall be attached to the car underframe in a visible location (see 6.6).

* 3.17 Painting and marking. Painting and marking shall conform to MIL-P-53044, color 14064, with these exceptions: (1) Stencilling shall be done with AAR approved white paint or other AAR approved material and must be of a consistency to thoroughly cover black background and provide a sharp contrast that will remain legible for a minimum period of 96 months. (2) Trucks shall be painted with a light application of TT-E-489 to attain 0.5 mils minimum dry film thickness that will not prevent detection of flaws or cracks in ordinary inspection. (3) Handgrabs and sill steps shall be painted safety yellow. (4) "JACK HERE" and "LIFT HERE" stencils shall be applied at the appropriate place. (5) The deck and side sills shall be stencilled to show the location of both the single and two tank configurations and their corresponding anchor locations. A color-coding system shall be used to clearly indicate the above for both the MI

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tank (white) and the M60 tank (yellow). (6) The deck shall be stencilled to show the correct pedestal locations for both 20 and 40 foot containers. Substitute systems affording equal protection are acceptable if approved by the Government. Reporting marks shall be DODX in lieu of USAX as specified in MIL-P-53044 and the words US Army shall not be applied. Stencilling on ends of car will include but not be limited to identification of couplers, wheels, end-of-car cushioning, springs and snubbers. Additional markings shall be in accordance with AAR's section L - Lettering and Marking of Cars, as applicable. Car numbers conforming to MIL-P-53044 shall be applied and shall be provided by the contracting officer. Decals shall be applied to the side sills when furnished (see 6.7).

3.17.1 Anti-skid deck covering compound. To the commercially sand-blasted steel deck plate, an epoxy primer conforming to MIL-P-52192 shall be applied in one coat of 1 to 1.5 mils thickness. Following this procedure, within a minimum of 4 hours and a maximum of 24 hours, an anti-skid deck covering compound conforming to specification DOD-C-24667, type IA, composition G, grade B, class 1 shall be applied with 3 to 5 mils thickness. All instructional markings to be applied to the deck surface thereafter shall be located only on areas that the tank treads will not traverse.

3.18 Lubrication. NFL roller bearings shall be grease lubricated in conformance to AAR requirements and applied in quantities approved by the bearing manufacturer. Truck centerplates shall be lubricated in accordance with AAR recommended practices.

3.19 Safety appliances. Safety appliances shall comply with the current issue of DOT Railway Safety Appliance Standards (Code of Federal Regulations, title 49, part 231). Four-inch wide sill steps conforming to AAR standards shall be provided to aid in safe operation of hand brake from side of car. Recessed design handholds shall be provided to ensure maximum deck width at ends of car.

3.20 Routing cardboard. An AAR approved routing cardboard shall be applied to each side sill near the car number.

3.21 Defect cardholder. An AAR type defect cardholder shall be applied to each car.

3.22 Workmanship. Workmanship for fabrication and construction shall be in accordance with AAR Section C, part II, volume 1 - Specifications for Design, Fabrication, and Construction of Freight Cars, M-1001.

3.22.1 Welding. Welding shall be in accordance with AWS D15.1 and AAR section C, part II. The surface of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing or other foreign matter. Spot, tack, or intermittent welds for strength will not be permitted. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided, when necessary, to reduce stress concentration.

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3.22.2 Welders. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall provide the Government with certification that the welder or welding operator has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ASME Boiler and Pressure Code, Section IX, Welding Qualifications.
AWS D15.1 Railroad Welding Specifications.

Contractors who only make horizontal welds need not qualify welders for "all position welding". In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator. The results shall be made available to the Government for review.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Acceptability criteria. Cars which conform to all requirements in sections 3 and 5 of this specification and pass all applicable examinations and tests in section 4 of this specification and meet requirements of the AAR Rules of Interchange will be considered acceptable by the Government. If the design is determined by the AAR to be of the "new and untried" type, the contractor shall furnish a letter of AAR interchange approval to the contracting officer.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards or for providing evidence that components have been previously approved by AAR.

4.2 Classification of inspections. The inspection requirements are classified as follows:

* a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

* 4.3 First article car inspection. The first article which shall have been produced from production tooling shall be examined and tested to determine conformance to the requirements of this specification.

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* 4.3.1 Examination. The first article car shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

* 4.3.2 Tests. The first article car shall be tested as specified in 4.5.2 through 4.5.2.7. Failure of any test or, if "new and untried", lack of AAR interchange approval shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each car shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each car shall be tested as specified in 4.5.2.2 through 4.5.2.5. The number of braking force tests (see 4.5.2.5) performed shall be in accordance with AAR requirements. Failure of any tests shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. Each car shall be examined for the following characteristics:

101. Material not as specified.
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of materials, material finishes, or treatment.
105. Parts and components not interchangeable.
106. Threaded parts not as specified.
107. Car dimensions exceed dimensions shown on AAR Plate B and B-1.
108. Dimensions not as specified.
109. Safety appliances not as specified.
110. Lubrication not as specified.
111. Identification plate not as specified.
112. Painting and marking not as specified.
113. Certification of construction and design approval not in accordance with AAR Specifications.
114. Workmanship not as specified.
115. Tolerances and fits are not as specified.
116. Brake equipment not as specified.
117. Defect cardholder not furnished or applied as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Any tests required by AAR shall be in accordance with AAR specifications, and as specified herein. No specific order of testing is required.

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4.5.2.2 Airbrake. The entire airbrake system shall be tested to insure correct functioning of all components. The car shall be tested with a single car testing device in accordance with AAR leaflet 5039-4, supplement 1, or latest revision for type of brake equipment. All pipe joints shall be tested for leaks at an air pressure of 90 psi. The brake cylinder piston travel shall be as specified in Air Brake specification no. 2518 and shall be tested for interference and correct travel. Nonconformance to 3.13 or AAR requirements shall constitute failure of this test.

4.5.2.3 Handbrake. The specified braking force shall be determined at each wheel. Nonconformance to 3.13.4 shall constitute failure of this test.

4.5.2.4 Side bearing clearance. With the car on level tangent track and the car body level, side bearing clearances shall be measured at each bearing. Nonconformance to AAR standard limits shall constitute failure of this test.

4.5.2.5 Braking force. Test in accordance with AAR requirements. Determine brake shoe pressure at shoe. Nonconformance to 3.13.2 shall constitute failure of this test.

4.5.2.6 Curvature. The car, uncoupled, shall be moved through a simulated 150 foot radius horizontal curve. At maximum swing, the brake rigging and the car body rigging shall be inspected to determine that when springs are at 75 percent deflection clearances have been provided and the brake systems function as specified. Calculations are to be furnished to prove that the car coupled to a locomotive by means of an auxiliary coupler similar to National 31629 will move through a 150-foot-radius curve. Nonconformance to this test or 3.7 shall constitute failure.

4.5.2.7 Cargo securement system. The cargo securement system's ability to restrain two M1 tanks and ANSI/ISO containers shall be demonstrated. Nonconformance to 3.11, 3.11.1, 3.11.2 and AAR requirements shall constitute failure of this test.

4.6 Inspection of packaging. The packaging and marking shall be inspected to determine conformance to section 5 of this specification.

5. PACKAGING

5.1 Car. Each car shall be prepared for delivery in accordance with AAR requirements for Interchange Service (see 6.8).

5.2 Marking. Marking for shipment shall be in accordance with MIL-STD-129. Additional marking shall be as required by applicable AAR rules and regulations.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The domestic service heavy duty flatcar is intended for transporting two tracked vehicles and a single 40-foot or three 20 foot ANSI/ISO containers or combinations of both.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2..1.1 and 2.2).

* 6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Data requirements. The contracting officer should include requirements for such data as Bill of Material; renewal parts catalogues; details of specialty items including appropriate part numbers, specifications, serial numbers, drawings and drawing numbers; Instructional materials; and Universal Machine Language Equipment Register (UMLER) Card and Document Input will be furnished as necessary.

6.5 Drawings. Drawings shall be made available to the contracting officer prior to submission of first article car. A complete set of blue line prints and black line reproducible drawings on 3 mil mylar shall be furnished to the contracting officer.

6.6 Car numbers. The contracting officer should arrange to furnish car numbers to the contractor to be applied to each car (see 3.16).

6.7 Decals. The contracting officer should arrange to furnish decals to the contractor when provided by the Military Traffic Management Command (see 3.17).

6.8 Packaging. Levels of packaging have not been included as the cars are for domestic service and will be delivered over the rails. Storage periods exceeding normal domestic limitations are not anticipated.

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* 6.9 Subject term (key word) listing.

Railway car
Railway car, flat
Railway car, heavy duty

* 6.10 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in theses notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army - ME

Review activity:
Army - MT

Preparing activity:
Army - ME

Project 2220-A205

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-52968D(ME)	Railway Car, Flat, 56-1/2-Inch Gage, Heavy Duty 140-Ton Capacity, Domestic Service

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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